

HANDBOOK



Use TECH To Create A New Idea That Changes The World

DATE: **7-9 FEB 2025**

VENUE: **Imperial Hotel,
Kuching Sarawak**

CATEGORIES

A

**PRIMARY GROUP
(SOFTWARE / HARDWARE)**

B

**SECONDARY GROUP
(SOFTWARE / HARDWARE)**

C

**TECH-HACKATHON
(PRIMARY & SECONDARY)**

D

**MICROBOTICS
CHALLENGE
(PRIMARY & SECONDARY)**

NEW IN 2025

*REMARKS: 7 - 18 YEARS OLD IN 2025



SCAN HERE



ORGANISED BY:

EDUWEL
Better Education, That's All We Do



ENDORSED BY:



OVERSEAS PARTNERS:





YTAC Event Overview

About YTAC

Since its inception in 2022, the Youth Tech Asia Challenge (YTAC) has become a premier platform for young tech enthusiasts across Asia to showcase their innovation and problem-solving skills. The competition brings together students with a passion for technology and aims to inspire the next generation of tech leaders. Over the years, YTAC has grown from a local event to an international platform, providing a space where students can demonstrate their creativity, technical skills, and collaboration abilities. In year 2025, a dynamic and innovative programming competition - Tech-Hackathon added designed to inspire and challenge young minds in the fields of technology and software development. Aimed at students and young enthusiasts, the hackathon provides a platform for participants to tackle real-world problems through coding, collaboration, and creativity. By combining teamwork with individual skill development, the hackathon encourages participants to think critically and develop practical solutions to everyday technical challenges.

Key Milestones of YTAC:

- **2022:** YTAC was launched in Kuala Lumpur, Singapore, Johor Bahru, Bangkok, and Penang, establishing itself as a regional event for youth to compete and exchange tech knowledge.
- **2023:** The competition shifted online due to global circumstances, allowing broader international participation and introducing new digital formats.
- **2024:** Hosted at Universiti Malaya, Kuala Lumpur, this year marked the first hybrid format, with both in-person and online participants from across Asia.
- **2025:** Now in its fourth year, YTAC continues to evolve with a fresh challenge format and new categories, particularly the Tech-Hackathon.

Our Mission

YTAC is more than just a competition. It serves as a platform for young minds to develop solutions to global issues using technology, fostering a collaborative and innovative mindset among students. Our core belief is that the youth hold the power to drive technological change and that through challenges like YTAC, they can cultivate the skills necessary to solve real-world problems.



Objective

- **Innovation:** Encouraging participants to think beyond conventional solutions.
- **Collaboration:** Fostering teamwork and communication in a diverse environment.
- **Practicality:** Ensuring projects have tangible real-world applications.
- **Inclusivity:** Offering a platform where students from all backgrounds can showcase their skills, regardless of location.
- **Technical skills:** Inspire participants to think critically about technical problems and develop solutions that could potentially address challenges in their local communities or on a global scale.



YTAC 2025 Overview

Event Details:

- Dates: 7th - 9th February 2025
- Venue: Imperial Hotel, Kuching, Sarawak
- Total team: 100 teams

Competition Registration Fee:

Free of Charge

Age Requirement:

All participants must be students aged between 7 to 18 years old in 2025.

Nationality Requirement:

The competition is open to students from all countries.

Team Size:

Each team must consist of 1 to 3 members.

Teams can be formed from the same school or organization, or from different schools.

Categories of YTAC 2025:

The grouping will be determined based on the **OLDEST** member's age in the team.

1. **Inno-Tech Primary Group:** aged 7 - 12 in 2025
 - Software
 - Hardware
2. **Inno-Tech Secondary Group:** aged 13 - 18 in 2025
 - Software
 - Hardware
3. **Tech-Hackathon:** aged 7 - 18 in 2025 (**NEW**)
 - Teams must 2-3 members
 - Solve real-world problems and prototype solutions within a limited time.
4. **Mikrobotics Challenge**
 - Teams must 2 members
 - Mission Challenge on map

Participant Format:

- **All Malaysian participants are required to attend the event in person** at Imperial Hotel, Kuching, Sarawak.
- **Online participation is ONLY available for non-Malaysian participants**, ensuring international participation and global collaboration.



Rules & Regulations

Category Inno-Tech Primary Group & Inno-Tech Secondary Group

Theme: Automation

1. Sub-theme:

- **Lifestyle & Security or Health or Transportation or Education**

2. New Projects:

- All entries must be new original tech projects, previously submitted projects or improved versions of past projects (i.e., “enchanted projects”) is prohibited.

3. Originality:

- All projects must be independently created by the participants or teams. Projects can fall under either the software or hardware categories and must include programming content and a complete presentation.

4. Programming & Technical Requirements:

- Projects must be developed using a programming language, and participants are free to use any type of programming language or technical tools.
- There are no limitations on the technical platforms used, but participants must ensure that the projects are original.

5. Submission of Files:

- Participants must submit their projects and all related documents online before the competition to confirm their eligibility. The submitted materials should include a roll-up banner design in PDF format and the project’s code file.

6. Content Restrictions:

- Projects must not contain any violent, explicit, political, racial, religious, or any other content that could disrupt social harmony. Any projects violating this rule will be disqualified.

7. Complete Presentation:

- Projects must be capable of being fully demonstrated during the competition, and participants must be able to clearly explain the project’s functionality, application scenarios, and the development process.

8. Other Requirements:

- Teams must comply with the organiser’s rules and regulations.
- All intellectual property rights of the projects belong to the participants, but the organiser reserves the right to display the projects during the competition.
- Any projects that are not submitted according to the rules or fail to meet the requirements will be disqualified.



Display Booth Setup & Requirements

(For Inno-Tech Category Only)

We will provide:

- 1 banquet table (2ft x 6ft) without a tablecloth
- 2 chairs

Participants are required to bring:

- X-banner
- Tablecloth
- Decoration materials
- Project display items (e.g., laptops, prototypes, or demonstration equipment)
- Any additional power cords or adapters required



For more information, please contact:
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X-banner Guidelines

Header:

- Project Title: A catchy and clear title for their project.
- Team Name: (if applicable) or Individual Name.
- Category: (e.g., Primary Group - Software / Hardware or Tech-Hackathon).

Main Content:

1. Problem Statement: A short and clear description of the problem the project aims to solve.
2. Objective: The purpose or goal of the project.
3. Key Features: Highlight 3-5 unique or standout features of the project.
4. Technology Used: Briefly list programming languages, software, or hardware used.
5. Target Audience/Users: Who benefits from this project? (e.g., students, businesses, communities).

Visuals:

- Diagrams/Images: Include schematics, screenshots, or photos of the project prototype.
- Flowcharts or Process Diagram: Illustrate how the project works (if relevant).
- Graphical Elements: Infographics or data visualizations showcasing results or impact.

Footer:

- Team Members' Names: (if team-based).
- Contact Information: (optional, for networking or queries).
- Acknowledgment: Mention support from mentors, schools, or organizations.

Additional Tips:

- Theme Alignment: Ensure the design reflects the theme of their project (e.g., eco-friendly, futuristic, etc.).
- Call to Action (Optional): E.g., "Discover more at our booth!"
- Logos: Include school logos or affiliations (if allowed).



Inno-Tech: Judging Criteria

Inno-Primary & Secondary Group

Projects will be evaluated based on the following criteria:

- Originality: Creativity and uniqueness of the idea.
- Technical Execution: Complexity and technical implementation.
- Functionality: Practicality and usability of the project.
- Presentation: The participant's ability to demonstrate and explain the project effectively.
- Peers Voting: Votes from the competitor teams' members.

***Golden Ticket** available: The extraordinary team will advance directly to the final stage in Hangzhou, China, subject to the judges' selection. Each judge will hold 1 Golden Ticket to award to their chosen team.*

Please register and submit your X-banner & code file by 31 January 2025, and fill out the related submission forms:

Submission Requirements:

- X-Banner: Upload a high-resolution design file.
- Code File: Submit all relevant project files (e.g., source code, documentation).
- Forms: <https://forms.gle/uLxgY7G25fPHwFJn6>

X-banner YTAC template:

<https://drive.google.com/drive/folders/1GMQdUf4-7UCbr8cYqwIDpuWES3m0P0O0?usp=sharing>



Rules & Regulations

Category: Tech-Hackathon Group

The Tech-Hackathon competition consists of two tiers: a solo competition (Tier 1) and a group competition (Tier 2).

Tier 1: Solo Competition (PVP Mode)

- **Overview:**
 - In Tier 1, individual participants compete in a series of problem-solving challenges. This solo part focuses on testing each participant's ability to resolve system bugs/errors and handle immediate technical issues under pressure.
 - Participants will compete using the Participant vs Participant (PVP) method, where competitors are paired randomly for head-to-head problem-solving matches.
- **Competition Rounds:**
 - The solo competition will proceed through several rounds, with participants facing off against one another until a final winner is determined.
 - Participants with the best performance, based on both problem-solving speed and accuracy, will be identified as potential MVPs of the Day.
- **Objectives:**
 - To evaluate individual technical skills, speed, and effectiveness in diagnosing and solving critical issues on the spot.
 - To identify the most skilled and quick-thinking individual participants.

Tier 2: Group Competition [Theme: Ordering Management System]

- **Overview:**
 - Tier 2 will focus on teamwork, where participants form groups and work collaboratively to solve a real-world problem within a given theme.
 - The theme for Tier 2 is "Ordering Management System". Participants can engage in self-training and pre-build exercises before the hackathon, but the actual subtopic and specific requirements will be revealed only on the competition day.
- **Scenario:**
 - The competition will simulate a real-world business scenario where a project is assigned to the participants by a business customer, investor, or boss.
 - Teams will have 5 hours to build an immediate, functional solution based on the assigned subtopic. They must work together to meet the outlined requirements and deliver a working system.



- **Expectations:**

- Projects must be original and developed entirely during the hackathon period.
- While teams can use pre-existing libraries and frameworks, the core functionality and solution must be developed during the event itself.

General:

3. Programming Language and Technical Rules

- **Backend:** Any backend programming language is allowed (e.g., Python, JavaScript, C++, C#).
- **Frontend:** Participants are free to use any frontend platform or framework to build their project.
- **Originality:** All projects must be original and created during the hackathon. Using templates or previously developed projects is strictly prohibited.
- **Plagiarism:** Plagiarism or direct copying from external sources will lead to immediate disqualification.

4. Mentor/Coach/Teacher Involvement

- **Limited Guidance:**

- Mentors, coaches, or teachers are not allowed to participate during the competition. However, participants can consult with their mentors briefly before the competition begins.
- Once the competition starts, participants are expected to work independently without external help.

- **Communication Restrictions:**

- Participants cannot use WhatsApp, Zoom, Google Meet, or any other telecommunication tool to seek help or guidance from outsiders during the competition.
- All work must be completed solely by the participating team members.

5. Code of Conduct

- **Professionalism:** Participants must maintain a high level of professionalism and respect toward other participants, judges, and mentors at all times.
- **Harassment and Cheating:** Any form of harassment, cheating, or violation of the rules will result in immediate disqualification.

6. Judging Criteria:

- Projects will be judged based on innovation, functionality, technical difficulty, and how well they meet the requirements set forth by the business scenario.
- **Time Management:** Teams must adhere to the allotted time limits. No extensions will be granted.
- **Team Collaboration:** All participants must contribute actively to the project; teams are expected to demonstrate effective collaboration.



Tech-Hackathon: Judging Criteria

Tech-Hackathon

- **Problem-Solving & Adaptability:** How creatively and efficiently the participant adapts to the problem and provides a solution within the time limit.
- **Technical Execution:** The complexity and accuracy of the solution, as well as the effective use of programming languages and tools.
- **Creativity & Innovation:** The originality of the idea and the introduction of inventive approaches to solve the problem.
- **Final Outcome & Demonstration:** The functionality, practicality, and clarity of the project in solving the problem.



Rules & Regulations

Category: Mikrobolic Challenge

1. General Requirements

1.1 The main challenge is for the Microbotic robot to move autonomously along a **BLACK LINE** on a track with intersections, passing through all scoring points from the start line to the finish line.

1.2 The robot must also achieve the **fastest time**.

1.3 The competition will be conducted physically, and Microbotic robots will be provided by the organizers, or participants may bring their own Microbotic robots.

1.4 The competition is open to all national and private secondary and primary school students aged 7 to 18 years.

1.5 Each team consists of a maximum of two (2) participants and one teacher advisor.

1.6 The teacher advisor is allowed to assist participants in checking the robot's functionality and performing test runs before the race.

1.7 The teacher advisor is not allowed to touch, repair, or code the robot during the race.

1.8 Any interference by the teacher advisor with the robot or jury decisions during the race may result in disqualification.

2. Robot Specifications: Autonomous Robot

2.1 Prepare a robot that can move autonomously along a black line (white background track) or a white line (black background track) on a circuit with intersections, passing through all scoring points (or as many as possible) from the **START** line to the **FINISH** line on the competition track.

2.2 Robots that cross the Finish line with the highest number of scoring points will have a higher ranking.

2.3 If scoring points are tied, the robot with the fastest time will be ranked higher and declared the winner.

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Rules & Regulations

Category: Mikrobotics Challenge

2.4 Robot Types:

2.4.1 Microbotic Robot.

2.4.2 Participants can use Microbotic robots purchased or provided by SSTP/USTP.

2.4.3 No modifications are allowed to the Microbotic robot's controller, sensors, output devices, or power source.

2.4.4 Must meet the specified construction, size, and weight requirements.

2.4.5 Can be programmed using text or graphical coding.

2.4.6 Coding can be done using the device library provided in the mBlock IDE software along with the Microbotic device.

2.4.7 Can be programmed using low-level coding with IDE software other than mBlock IDE.

2.5 Size and Weight:

2.5.1 Construction: Microbotic Robot.

2.5.2 Weight: No limit.

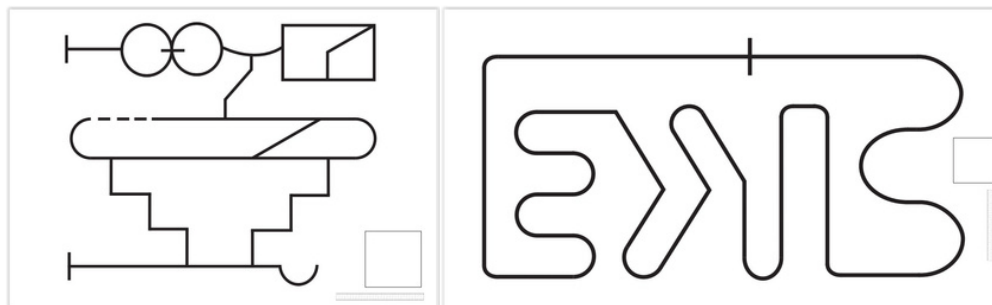
2.5.3 Width: Maximum 150 mm.

2.5.4 Length: Maximum 150 mm.

2.5.5 Height: No limit.

2.5.6 Inspection: The robot will be inspected during the race.

Mission Maps:



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YTAC 2025 Rundown

Day	Agenda
07.02.2025 (Friday) 2pm - 5pm	• Tech-Hackthon Workshop • Mirobotic Workshop
08.02.2025 (Saturday) 9am - 5pm	• Opening Ceremony • Setup for Challenge • Lunch • Showcase & Judging • Educators Forum • Day End
09.02.2025 (Sunday) 9am - 5pm	• Peers Voting & Exchange • Final Result Calculation • Lunch • Closing & Awards Ceremony • Photo Session • Day End



YTAC Road to Glory 2025

100 teams



Youth Tech Asia Challenge (YTAC)

Kuching, Sarawak

(7-9 Feb 2025)



Top 30 teams



**International Creativity & Innovation
Awards (ICIA)**

Vietnam

(25-27 Apr 2025)



Final 20 teams



**International Innovative Tech Summits
(IITS)**

China

(TBC, Nov 2025)

